

ORIGINAL RESEARCH

Evaluating State Nitrification Action Plans for Controlling Distribution System Nitrification and MDBPs

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ABSTRACT

With public water systems switching to chloramination for secondary disinfection, distribution system nitrification and diminished disinfectant residuals are a growing concern. While some states have adopted Nitrification Action Plans (NAPs), no comparative assessment exists on their impact on drinking water quality trends. This study uses causal impact analysis to evaluate the effectiveness of state-mandated NAPs in controlling nitrification. Chloraminated systems with residual challenges before NAPs showed increased residuals and at least a 26% reduction in total coliform detection post-mandate. Maintaining chloramine residuals above 1.5 mg/L can prevent nitrification and reduce violations without significantly increasing disinfection byproduct occurrences. In contrast, states with no notable challenges before the plan's enactment saw limited benefits. These findings offer policymakers valuable insights into the plan's implications, supporting informed decision-making, especially in areas where nitrate violation hotspots exist without a current plan.

1 | Introduction

Chloramine has been used in the United States since the early 1920s to maintain disinfectant residual in drinking water distribution systems (AWWA 2013). Although its usage may have decreased during World War II due to ammonia shortages, many systems have renewed their interest in chloramination primarily to comply with the disinfection byproduct regulation. After the EPA introduced the Disinfectant Byproduct (DBP) Rules, over 29% of community water systems switched from free chlorine to chloramine disinfection, with two-thirds of medium and large systems experiencing nitrification issues to some degree (Seidel et al. 2005; Khiari 2019). According to disinfection survey results, many states continue to anticipate smaller systems to switch to chloramines as a solution to disinfection byproduct issues (AWWA 2018). Although switching to chloramine could

reduce the formation of DBPs, this can create a new burden on many systems that are unfamiliar with what the ramifications will be post transition. There have been concerns of distribution system nitrification caused by ammonia addition for chloramine formation and subsequent chloramine decay. Uncontrolled nitrification can lead to disinfectant residual loss, unexpected rises in nitrite and nitrate levels from ammonia oxidation, microbial growth, and potential release of lead and copper from pipes (Hossain et al. 2022; Khiari 2019). Concerns about distribution system nitrification and adverse water quality impacts have proven to be the biggest deterrent against switching and a major challenge for utilities relying on chloramine for disinfection.

The drinking water distribution system is a component in the water system that carries drinking water from the treatment plant to consumers' taps (Figure 1). It is a critical control

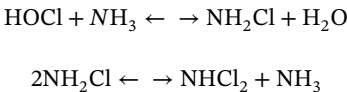
Article Impact Statement

State nitrification action plans help chloramine utilities maintain residual levels, reduce total coliforms, and pro-actively manage nitrification without significantly in-creasing MDBP occurrences.

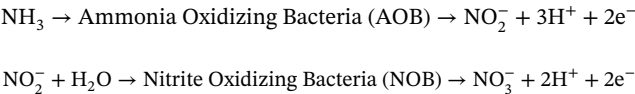
point for safe drinking water and therefore requires proper operation and maintenance. To ensure optimal quality and public health protection, the current Safe Drinking Water Act (SDWA) includes five regulations that apply to the distribution system: the Surface Water Treatment Rules (SWTR), Stage 1 and 2 Disinfectant and Disinfection Byproduct Rule (DBPR), Ground Water Rule (GWR), Revised Total Coliform Rule (RTCR), and the Lead and Copper Rule (LCR) (Smith and Slabaugh 2017). These regulations focus on water quality but do not specifically address nitrification and monitoring requirements within the distribution system outside of sanitary surveys. Under chloramine chemistry, particularly in the monochloramine zone of a breakpoint chlorination curve (at a $\text{Cl}_2\text{:NH}_3\text{--N}$ ratio between 5:1 to 8:1 by weight), the decay of monochloramine (NH_2Cl) can release free ammonia (NH_3), which serves as a substrate for nitrifying microorganisms (Bradley et al. 2020).

The series of reactions that drive this process are shown below. These reactions illustrate how excess free ammonia can increase the risk of nitrification. To minimize this risk, ammonia should be dosed based on chlorine residuals; specifically, chlorine should be measured immediately before ammonia is added to prevent overfeeding. Based on the stoichiometric balance of the chloramine reactions below, the target ratio of chlorine to ammonia ($\text{Cl}_2\text{:NH}_3\text{--N}$) is approximately 1:1 by mole or roughly 5:1 by weight (i.e., 70mg/mmol Cl_2 per 14mg/mmol $\text{NH}_3\text{--N}$). Therefore, many well-developed Nitrification Action Plan (NAP) recommends this dosing strategy to help operators minimize nitrification risks, along with minimum monitoring requirements, as summarized in Figure 1.

Chloramine Formation and Degradation:



Nitrification Pathway:



Nitrite is a key indicator of nitrification and may increase during the initial stage (AOB) or decrease as nitrite gets converted to nitrate (NOB). An increase in nitrate often indicates severe nitrification. Numerous studies have shown that excess nitrate and nitrite in drinking water can cause methemoglobinemia (blue baby syndrome) and may be associated with certain cancers, birth defects, and thyroid issues (Pennino et al. 2017). The current maximum contaminant level (MCL), according to SDWA, for both nitrate and combined nitrate-nitrite in drinking water is 10mg/L-N, and for nitrite is 1mg/L-N, regulated at the entry point. There are currently no regulations or MCLs for nitrate and nitrite within the distribution system. If these MCLs were applied to the water supplies at the tap, it is possible that many systems would exceed these thresholds when nitrification occurs and may also lead to violations in the SWTR and Total Coliform Rule (TCR) due to the increased microbial activity, disinfectant residual loss, and total coliform growth. Furthermore, reduction of pH and alkalinity from nitrification may result in the release of lead and copper from pipes, impacting LCR compliance (AWWA 2013). While most states follow exactly the federal mandate developed by EPA to maintain compliance, some states have stepped in with their own regulation requiring community water systems (CWS) to have a NAP.

We define NAPs as state-enforced, site-specific monitoring and control plans designed to ensure that chloramine disinfection is successful by preventing and/or responding to nitrification within public drinking water distribution systems. For this

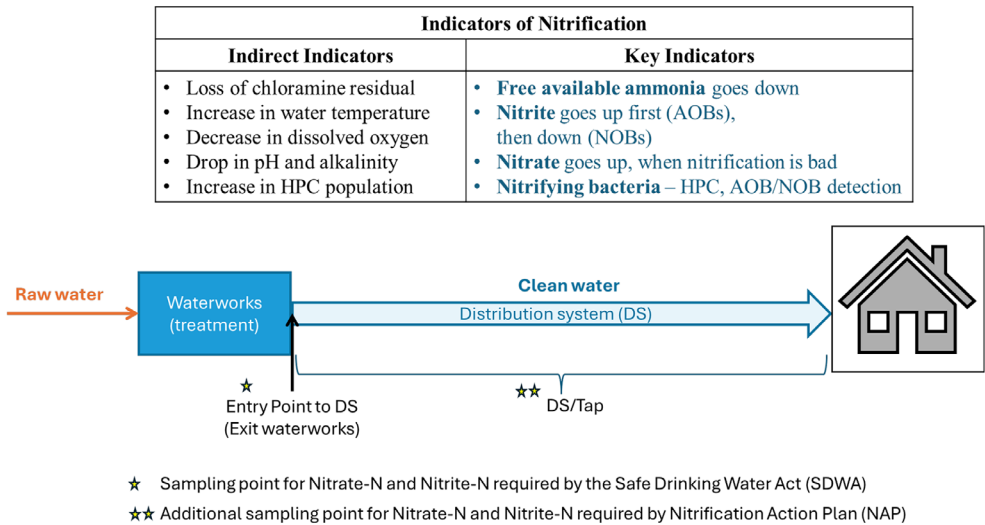


FIGURE 1 | Schematic of a water supply system highlighting potential indicators of nitrification within the distribution system. Adapted from (TCEQ 2023).

study, a system was considered to have a Nitrification Action Plan if it operated in a state with a statewide NAP mandate requiring utilities to: (1) establish site-specific goals; (2) define clear baseline and trigger levels for disinfectant residuals, ammonia, nitrogen, and microbial indicators within the distribution system; and (3) maintain a well-documented nitrification response plan in alignment with the AWWA M56 manual.

There are several contributions this study makes. First, it is the first study to comparatively analyze states that NAP mandated states with those lacking such requirements (Non-NAP states) and statistically evaluate the impact of nitrification control plans on disinfectant residual concentrations, total coliform, nitrate/nitrite concentrations, DBPs, lead violations, and copper violations. An effective NAP would not only ensure higher residual post implementation but also minimize associated trade-offs. Second, by interviewing state regulators, this research contributes for the first time a better understanding of what a typical plan consists of, and the reasons and challenges involved in plan development from an operator and regulator's point of view. Finally, by using a controlled time series and multilevel intervention analysis to evaluate the impact of NAP, this paper opens several avenues for future research, including how challenges of nitrification and residual loss might be driven by multiple distribution system characteristics and how control strategies can be optimized to protect public health.

2 | Methodology

2.1 | Data Sources

Since nitrate and nitrite monitoring are not required at the distribution system, there currently does not exist a database of distribution system concentrations to directly investigate this problem; therefore, this analysis relied on a short survey, publicly available databases, and additional state-managed data for analysis. This study compiled water quality data from multiple databases, focusing on states within EPA Region 6 where the number of chloraminated systems, the potential for nitrification due to contaminated source water, and microbial concerns are highest, as documented in regulatory and industry reports (Ward and Sherchan 2023; Yoder et al. 2010; Pennino et al. 2017; EPA 2024b; Hall et al. 2024). The data were selected for periods between January 1, 2009 and January 1, 2019 for Louisiana's case study, and between January 1, 2010 and January 1, 2020 for Texas' case study. Only public water systems (PWSs) that consistently reported either chlorine or chloramine residuals throughout the study period (2006–2022) were included; systems that switched disinfectant type were excluded (see Figures S7 and S8). The final dataset includes 5608 unique PWSs: 162 in Louisiana, 3606 in Texas, 714 in Arkansas, 246 in New Mexico, 861 in Oklahoma, and 19 tribal systems. The available databases allowed for the assessment of disinfectant residual concentrations, total coliform detection, DBPs occurrences and concentrations, and nitrate-nitrite violations from 5 years before and 5 years after NAP implementation and explored the benefits and potential trade-offs of state-mandated NAPs. Contaminant violation and occurrence records from the EPA's database were queried

iteratively and processed using R statistical software (R Core Team 2023).

2.1.1 | State Regulator Survey

An online Qualtrics survey was distributed to each state by email, with 31 total responses received and reviewed for completeness. This allowed for the identification of states within the continental U.S. (CONUS) with a known State-enforced NAP. Details of the 25-question survey questions used in this study are provided in the SI Methodology: State Survey section. Responses from the survey were used to identify states in the CONUS with NAPs, states without NAPs, and states with no known response. The NAP mandate is treated as the “treatment” or intervention, with NAP states assigned to the “Treatment Group” and Non-NAP states to the “Control Group.”

2.1.2 | EPA'S Safe Drinking Water Information System

Since the SDWA enactment in 1975, PWS have been reporting violations to the EPA Safety Drinking Water Information System (EPA 2024b). Safe Drinking Water Information System (SDWIS) provides publicly available violation data for over 90 contaminants, including total coliform, disinfection byproducts, heavy metals, radionuclides, inorganic chemicals, nitrite, and nitrate. Using the year and quarter the violation occurred, and information on PWS water sources type, system type, sampling point, and population serving size, a variety of metrics were computed to compare between chloramine utilities with NAPs and without NAPs. The study examined each state's violation rates over a 10-year period, marking the state's NAP enforcement date, to see if implementation of NAPs helped reduce the number of systems in violation for each contaminant. The analysis was restricted to active CWSs because these systems serve the same customers year-round. All water sources, both groundwater and surface water, and both wholesaler and purchased systems were included. The violation data was used to compare the number of PWSs experiencing nitrate rule, total coliform rule (TCR/RTCR), lead and copper (LCR), and disinfection byproduct rule (DBPR) MCL violations during the period before and after implementation of State NAPs. Both Average (02) and Single Sample (01) MCL violation codes were selected for MCL violation data, and Level 1 (LV1A) and Level 2 (LV2A) Treatment Technique (TT) triggers were included for the assessment of RTCR violations.

2.1.3 | Six-Year Review (6YR)

The EPA Six Year Review (6YR) contains occurrence and contaminant level data for each public water system since 1996 and provides a report on regulated contaminants at least once every 6 years (EPA 2023). The 6YR3 and 6YR4 data were used to compare disinfectant residual level and contaminant occurrence data for total coliform, haloacetic acid (HAA5), and total trihalomethanes (TTHMs). Large systems may have multiple sampling locations; therefore, to standardize the results, a locational running average (LRA) of each contaminant for each PWS was calculated without considering sample location (Samson and Seidel 2022).

Since different PWS types may have different sampling methods, this study focused on only active CWSs with all water sources, both groundwater and surface water, and both wholesaler and purchased systems included for the analysis. A systematic approach was used to determine a cutoff value, beyond which values are considered outliers due to misreported units. For example, 4 mg/L-Cl₂ was used as a reasonable upper-bound limit for all distribution system residuals. Thus, any residual value above the maximum residual disinfection level (MRDL) of 4 mg/L-Cl₂ in the distribution system would be considered an outlier and either transformed into the correct units or excluded as necessary.

UCMR4 facility data were used to relate 6YR monitoring data to facility characteristics including system size, the primary source water type, sampling point, and the secondary disinfectant type (EPA 2024a). Unfortunately, the UCMR characterizations of secondary disinfection practices were only available for 4963 PWSs and only became available in UCMR4 (2017–2019) and onwards. Therefore, secondary disinfection practices for public drinking water systems prior to 2017 were considered incomplete and inconclusive. To close this data gap, we used the assumption that systems in the 6YR which report disinfectant residual concentrations as free chlorine use chlorine disinfectant and systems, which report residual as total chlorine use chloramine. This classification method was necessary to make possible the assessment of total coliform and DBP occurrence data as a function of treatment or disinfection practices over any period. The assumption was validated by cross-referencing disinfection classification with UCMR4 and state provided PWSID/Treatment database, showing > 75% agreement.

2.1.4 | State-Managed Compliance Data

State-managed compliance data were collected from two states: Texas and Louisiana, for dates between 2009 and 2020. For the case study, Texas was chosen to represent a state with high residual prior to NAP (> 1.5 mg/L pre-intervention), and Louisiana was selected to represent states with low residual (< 1.5 mg/L pre-intervention). The datasets from each state were collected through direct requests to the state regulatory agencies. Datasets include occurrence and concentration data as well as PWS inventory, purchasing relationship, and disinfection type history to determine the number of systems active and practicing chloramine disinfection with NAPs. A thorough quality check (QC) was performed on each state dataset, and the datasets were reformatted to a consistent format and compiled together in one database. The state-managed compliance data utilized in this study include TC, TTHM, and HAA5 occurrence and concentration data in the distribution system. The purchaser and seller information provided was used to link purchased water systems that report nitrate-nitrite concentrations at the entry point (EP) to their sellers; monitoring data from these purchasers were then represented as Distribution System (DS) nitrate-nitrite samples for the public water system selling the water.

2.1.5 | Nitrate–Nitrite Data

There is no national database for nitrate and nitrite data beyond the entry point to the distribution system. Thus, nitrate and

nitrite analysis relied solely on data acquired through a combination of email requests and the SDWIS database, each with limited reporting and its own set of limitations. A majority of PWSs only had started monitoring for distribution system nitrate and nitrite after the state mandated NAP date, and limited data were collected before the date for comparison. Meanwhile, the SDWIS database only reports data for systems in violation, coupled with contaminant concentrations that are restricted only to those above the MCL.

2.2 | Data Aggregation and Analysis

Distribution system water quality can be affected by seasonal changes in source water quality or by changing amounts of disinfectant residual, which must be accounted for. Daily, monthly, and yearly averages were considered. To test for stationarity and seasonality, the Autocorrelation Function (ACF) plot and Partial Autocorrelation Function (PACF) plot were used, and the best Autoregressive Integrated Moving Average (ARIMA) model was selected using the Bayesian information criterion (BIC). Final disinfectant residual, HAA5, TTHM, and frequency of TC detection data were aggregated as quarterly locational running annual average (LRAA) over a 10-year period (5-year pre and 5-year post NAP intervention) for the analysis. An overall trendline was added to the time series plot for both observed and unobserved (counterfactual) datapoints to remove quarterly seasonality as validated by the ACF/PACF plot, as discussed in Figure S3. This method strikes a balance between noise reduction, trend visibility, and insight.

To evaluate the effects of NAP implementation on each contaminant, we used a two-part strategy. First, we applied Interrupted Time Series (ITS) analysis using ARIMA(0,0,0)(0,1,0) to model the counterfactual and allow us to estimate both the immediate and longer-term effects of the intervention without any control group. Second, we used a Difference-in-Differences (DID) framework, which requires a control group to account for other time-varying factors that could confound the results. These factors would include simultaneous improvements in treatment practices triggered by separate issues such as responses to *Naegleria fowleri* detections, elevated disinfection byproducts, efforts to reduce total coliform, or internal motivation by water suppliers to improve system performance. The ITS model does not use a control group but allows us to visualize temporal changes, such as shifts by quarter and year (as applied here). We paired this with a DID model, which estimates the average effect of NAP implementation by comparing NAP to Non-NAP systems before and after the state mandate (O'Neill et al. 2016; Li et al. 2021). Together, they provide a more complete picture of both trends and net changes. Comparisons between NAP and non-NAP systems were limited to community water systems (CWSs) within EPA Region 6 that had a clearly defined NAP implementation date. These systems were selected because they operate under similar regional regulations, climatic conditions, source water characteristics, and treatment practices. Slope estimates from the ITS (Equations 1 and 2) and DID models (Equations 3 and 4) were used to assess both immediate and long-term impacts on contaminant concentrations and detection rates. The ITS model has the following structure shown in Equations (1) and (2):

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 [t - T_t] D_t + \varepsilon_t \quad (1)$$

where Y_t : outcome at t timepoint (e.g., quarterly average concentrations or probability of positive detects). β_0 : baseline average (intercept) *before* the intervention. t : time (in quarters). T_t : time point when the intervention (NAP) occurred. β_1 : pre-intervention trend (slope before NAP). D_t : dummy variable: 0 for pre, 1 for post intervention period. β_2 : immediate level change *after* NAP implementation. β_3 : change in trend after NAP (i.e., difference in slope post-intervention). $[t - T_t] D_t$: total time elapsed since intervention, only active in the post-intervention period. ε_t : error term, accounting for deviations from the fitted model (can include autocorrelation or heteroskedasticity).

The equation treats the implementation of NAPs as a *treatment*, establishes a *baseline* for all states, and then compares those with NAPs to those without NAPs to evaluate the *difference-in-difference* treatment effects. The equation above can be translated to:

$$Y_t = \beta_0 + \beta_1 \text{Time} + \beta_2 \text{Intervention} + \beta_3 (\text{TimeSinceIntervention} \times D_t) + \varepsilon_t \quad (2)$$

This regression highlights both the immediate (β_2) and long-term (β_3) effects of NAP implementation on chloramine residual, total coliform detection, and DBP occurrence. The same dataset used for ITS was used for DID analysis, which contains PWSs that undergo NAP intervention (Treatment Group) and PWSs that did not undergo NAP intervention (Control Group). Each group was registered pre- and post-treatment outcomes. Each coefficient represents a group mean or a difference between two means, following Equations (3) and (4).

The DID model has the following structure:

$$Y_t = \beta_0 + \beta_1 T + \beta_2 D_t + \beta_3 T \times D_t + \varepsilon_t \quad (3)$$

where Y_t : outcome at t^{th} timepoint (e.g., quarterly average concentrations or probability of positive detects). β_0 : average outcome of the control group *before* the treatment. T : dummy variable indicating treatment whether a group is the treatment group ($T=1$) or control group ($T=0$). β_1 : baseline difference between the treatment and control groups before the intervention. D_t : dummy variable indicating whether the observation is from *before* (pre=0) or *after* (post=1) the intervention. β_2 : captures the *change over time* in the control group (i.e., how the outcome changes in the post-intervention period, independent of the treatment). β_3 : key parameter of interest. It represents the additional change in the outcome for the treatment group after the intervention, beyond what would have occurred without the treatment. This is the *difference-in-differences estimate*. If $\beta_3 = 0$, it can be concluded that the treatment (i.e., NAP) had no effect. $T \times D_t$: equals 1 only for treated units after the intervention, allowing us to isolate the treatment effect.

Welch two-sample t -test statistics were used to determine if there is a significant difference between mean values pre- and post-NAP intervention period. When interpreting the p value of adjusted results from the DID analysis, it is crucial to take into account any differences in sample size, as these variations can impact the significance of the p value (Karpen 2017;

Andrade 2019). It is important not to solely rely on the p value from DID for inference. The equation above can be translated to Equation (4).

$$Y_t = \beta_0 + \beta_1 \text{Treatment} + \beta_2 \text{Post} + \beta_3 (\text{Treatment} \times \text{Post}) + \varepsilon_t \quad (4)$$

As illustrated in Figure 2, two differences were taken. First, data before the NAP intervention was subtracted from the data after the intervention, for both treatment and control. The treatment differences contain the NAP effect of interest and other effects that may not be of interest. The control differences contain only those effects of no interest. Second, these two differences are subtracted to remove the impact of the factor not of interest and potentially isolate the causal effect of the NAP impact.

While we expect DID and ITS models to adequately estimate NAP effects, we also implemented a multilevel (hierarchical) model (MLM) to account for potential bias arising from unobserved differences across public water systems (PWSs) throughout the study region. Free chlorine systems, which are not expected to respond to NAPs, were included as a negative control to adjust for external trends unrelated to policy exposure. To build the MLM, we specified fixed effects for key interactions (e.g., time, policy, policy group, time since intervention, and disinfection type), and included random intercepts (e.g., PWSID) to account for system-level variation. We developed and compared a set of candidate models ($n=8$), evaluating model fit using Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and intraclass correlation coefficients (ICCs) (Gelman and Hill 2021). The model with the lowest BIC was selected for further analysis.

3 | Results and Discussion

3.1 | Identifying States With Nitrification Action Plans and Common Monitoring Elements

To identify states that currently have nitrification plans, a survey was conducted and distributed to state regulators. The total of 31 responses, mapped in Figure 3, allows for identification of states with enforceable guidelines and the year that they were enacted. Out of 49 states in the COTUS, there are currently 6 states (12%) with enforceable NAPs, 25 states (51%) without enforceable NAPs, and 18 states (37%) whose status is undetermined at the time of this study.

At least five states, Texas, Iowa, Louisiana, Pennsylvania, and Illinois, have required utilities to have a nitrification action plan. Important dates and criteria governing the NAP for these states are shown in Table 1. Massachusetts has a plan in effect but has not yet made it mandatory. Minnesota and Kentucky are in the process of plan development and therefore are not included in Table 1. State regulators in Kentucky are currently in discussion about making a NAP a requirement at the time of this study. States without current plans enforced or in development fall into one of a few categories: those that expect NAP requirements in the future (New Hampshire, Oklahoma, Kansas, Kentucky, Massachusetts); those without any (New Mexico, Idaho, Alabama) or with few (Washington, Oregon, Vermont)

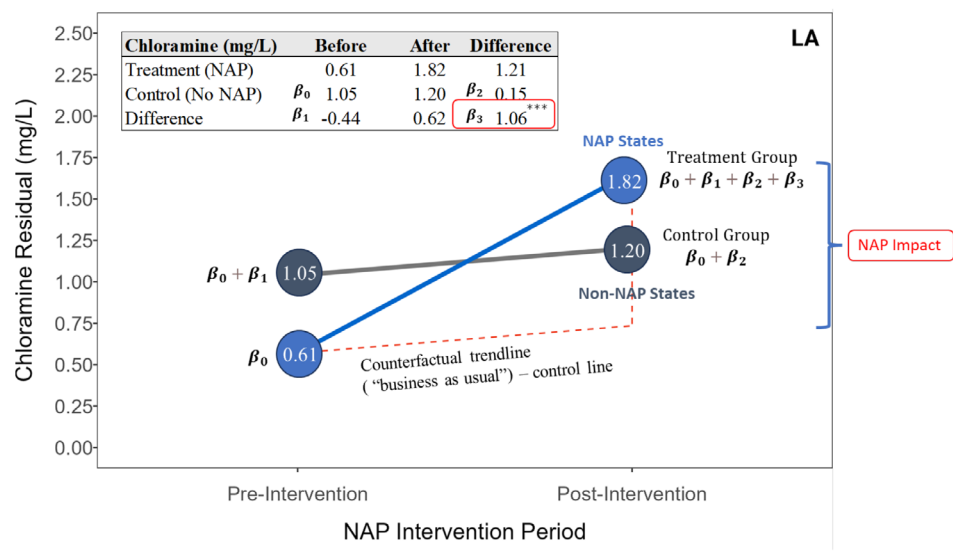


FIGURE 2 | Example DID showing significant causal effect of NAP on Louisiana's mean chloramine residual value measured as mg/L-Cl₂.

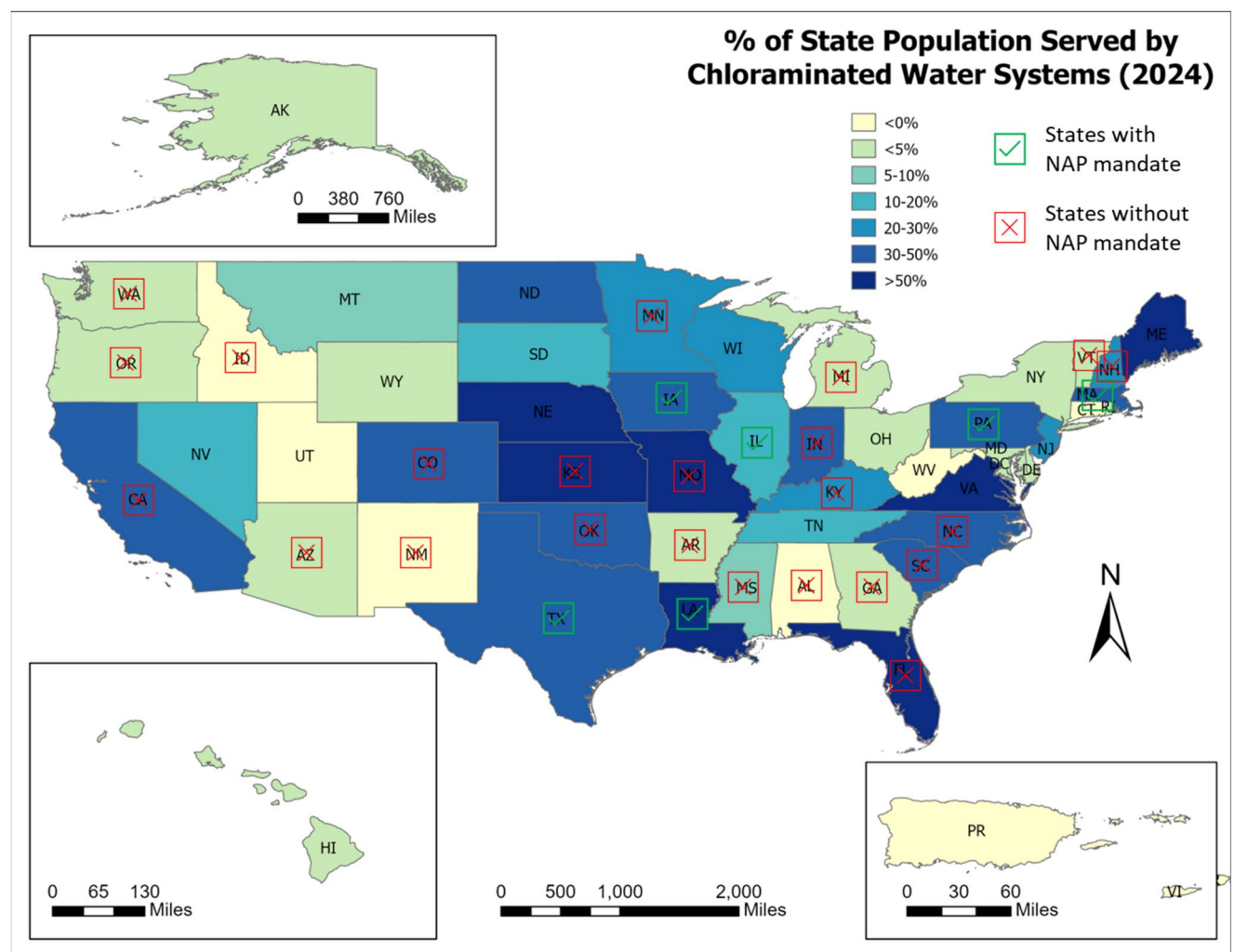


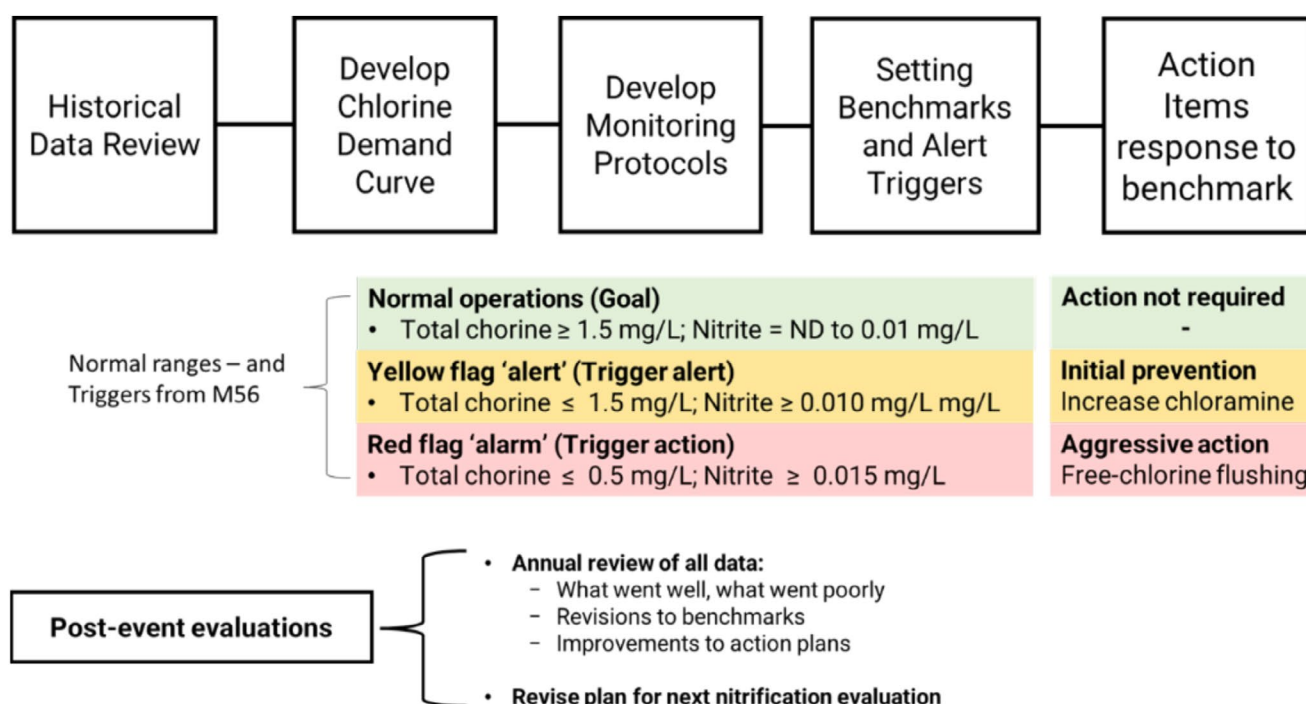
FIGURE 3 | 2024 Map of drinking water treated with chloramine across the United States and their state-mandated nitrification action plan status reported according to 2023 survey results (see Table S1).

chloramine-disinfecting utilities; those with few nitrification issues (Colorado) typically due to cold temperature or no TOC; those with operational and management concerns such as

having mainly small systems (AZ) or experiencing widespread workforce shortages (IN); those that are already proactive with an effective monitoring system in place (RI, SC, VT); or those

TABLE 1 | Survey responses for states with an effective Nitrification Action Plan.

State	What criteria are required?	What is the enforcement year for these plans?	By what year did the state require utilities to have NAPs?	Administrative code
Texas	30 TAC §290.46(z)	2015	2015	Title 30, Texas. Administrative Code (30 TAC) Chapter 290
Iowa	State provide template	2016	2016	Iowa Administrative Code 567—Chapters 43
Louisiana	AWWA M56 Manual	2014	2017	Louisiana Administrative Code, Chapter 3, Section XII-367
Pennsylvania	AWWA M56 Manual	2019	2019	25 Pa. Code § 109.716
Illinois	State provide template	2019	2019	Title 35, IL Adm Code Section 604.140
Massachusetts	Plan development phase	In effect	No projection	NA

**FIGURE 4** | Flowchart overview of a typical NAP development process. Utilities initiate a NAP in response to recurring signs of nitrification or elevated chloramine decay, guided by best practices such as those outlined in the AWWA Manual M56—*Nitrification, Prevention and Control in Drinking Water*. Action triggers may include declining chloramine residuals, increasing nitrite/nitrate levels, rising HPC or coliform counts.

that do not feel the need due to lack of a regulatory driver (GA, NC, MO). California did not fit into these categories. However, if NAPs were to be required, state regulators indicated that monitoring requirements would likely be tailored to each individual utility.

According to the survey, a common element of a Nitrification Action Plan for each state includes routine monitoring, establishing “baselines,” and setting “goals” for total and free chlorine residuals, free ammonia-N, heterotrophic plate count, nitrite-N, and nitrate-N to ensure disinfection is maintained

correctly (Figure 4). Most utilities with NAPs monitor entry-point nitrite and nitrate at least monthly to set baselines, then quarterly. In the distribution system, nitrite and nitrate are monitored at least quarterly and in response to trigger events, while total chlorine, free ammonia, and chloramine are typically monitored weekly. At minimum, inclusion required (1) sampling to make sure the correct chlorine-to-ammonia ($\text{Cl}_2:\text{NH}_3\text{-N}$) ratio exists (i.e., maintaining a desired range of $\text{Cl}_2:\text{NH}_3\text{-N}$ of ~4:1–5:1), (2) maintaining a baseline total chlorine residual of 0.5 mg/L of Cl_2 throughout the distribution system with action triggers, and (3) monitoring any changes in

nitrate and nitrite. The EPA's AWWA M56 Manual and guidance from the Texas Commission on Environmental Quality (TCEQ) offer valuable resources for developing Nitrification Action Plans and can serve as key references for utilities. Once baselines are set, deviations from baseline trends can signal the onset of nitrification. The definition of a "deviation" varies by utility and is determined in coordination with the state primacy agency as part of each utility's NAP. A total chlorine residual level of $>1.5\text{--}2.0\text{ mg/L-Cl}_2$ is typically recommended by the state for reducing risk of nitrification with $<1.5\text{ mg/L-Cl}_2$ level set as action triggers (AWWA 2013); therefore, increased disinfectant residual throughout the distribution system is expected as a NAP enactment impact.

For PWSs with NAPs, routine sampling was used to detect potential nitrification with triggers such as "yellow flag" alerts to indicate when values deviate from the norm, and "red flag" alarms when there is a strong possibility of nitrification requiring immediate action to get the system back to normal. Based on the severity of the suspected nitrification, typical corrective actions in response to nitrification may include, but are not limited to, increasing entry-point and distribution total chlorine residual levels, reducing free ammonia levels, water main and/or unidirectional flushing, replacing aged pipes, performing periodic free chlorine conversion, water storage maintenance, and pH adjustments to return the system back to normal operation (TCEQ 2023). The criteria outlined in AWWA Manual 56 were commonly used as a framework to develop templates distributed to PWSs. The survey responses provided valuable insights into the challenges that must be taken into consideration (from the states' perspectives) in the enactment of these plans to include (1) having a limited number of chloramination utilities to not warrant a statewide NAP, (2) the belief that NAPs do not offer a universal solution to water quality issues, and (3) the challenge of educating operators on the complex chlorine chemistry needed for effective NAP implementation.

3.2 | Temporal Trends for Distribution System Violations

To evaluate the impact of state NAPs on total coliform (TC), lead and copper rule (LCR), and DBP (HAA5 and TTHMs) violations within the distribution system, data from SDWIS was processed and analyzed. Higher residual concentrations can prevent nitrification and bacterial growth in the distribution system, contributing to the observed reduction in TC violations after the NAP implementation date. While this decline may also reflect broader regulatory effects, it is possible that NAP may play a contributing role, as the decline in TC violations begins immediately after the NAP implementation date (i.e., after 2014 for Louisiana and after 2015 for Texas). Figure 5 presents MCL violation trends for each contaminant to examine these trade-offs. This observation will be explored further in subsequent analyses; higher residuals may also increase the risk of disinfection byproduct formation and lead and copper violations.

One limitation when analyzing violation data is that monitoring is inconsistent across the states and there is a potential for underreporting violations, with estimations suggesting that between 26% and 38% of health-based violations are either

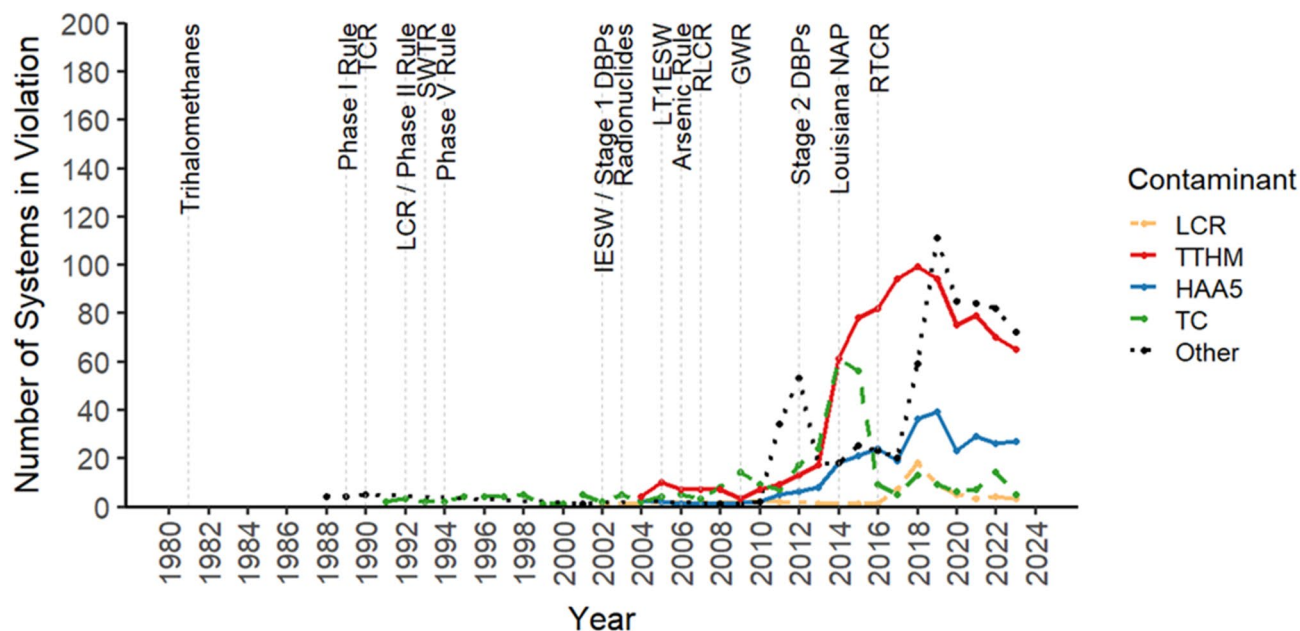
unreported or inaccurately documented (40 CFR 141.2 1975; Allaire et al. 2018). Another limitation is the lack of reported nitrate and nitrite violations within the distribution system, making it difficult to draw definitive conclusions from entry-point nitrate and nitrite violations reported in SDWIS. At least 45 systems in Louisiana were documented to have switched disinfectants during the study period (see Figure S7). In contrast, historical records from Texas indicate that between 2011 and 2022, over 400 community water systems (CWSs) transitioned from free chlorine to chloramine disinfection (see Figure S8). Analysis on Texas' violation data showed that the switch to chloramine was followed by an increase in LCR violations between the years 2011 and 2014. Following the state's NAP date in 2015, fewer TC, HAA5, and TTHM violations were observed, while the number of LCR violations slightly reduced. This suggests that the disinfection switch to chloramines may have had much more influence on Texas' LCR (Figure 5B) and possibly nitrite violations (Table 2) when compared to the effects of a NAP. Further analysis is needed to draw conclusions using only data restricted to systems with consistent disinfection types and excluding those that switched, which may require access to state internally managed data.

3.3 | Changes in Contaminant Levels Following Nitrification Action Plan Implementation

Analysis of NAPs on contaminant concentrations focused on EPA region 6, comparing three states without NAPs—New Mexico, Oklahoma, and Arkansas—against two states with statewide NAP implementation, Texas and Louisiana. It is important to note the specific timelines of NAP enactment in these states shown in Table 1. In Louisiana, a voluntary NAP was initially implemented in 2014, followed by a mandatory update in 2017 requiring compliance from all systems. Texas both implemented and mandated a NAP in 2015.

Regression statistics in Figure 6 are based on a non-parametric test, the Mann-Whitney *U* test (Wilcoxon rank-sum test), to compare the median disinfectant residual values between the pre- and post-NAP periods. This was based on a balanced random sample ($n=30$ per group) to avoid inflated *p* values from large samples (Lin et al. 2013; Karpen 2017). Significance levels are denoted as $p < 0.001$ (***), $p < 0.01$ (**), $p < 0.05$ (*), and not significant (ns). Non-NAP states (Figure 6A) showed no significant change in disinfection residual values pre- and post-NAP. In contrast, analysis on Louisiana's NAP (Figure 6B) revealed substantial increases in disinfection residual concentrations for both free chlorine and chloramine systems. The increase was more notable in chloramine systems (+198% from 0.30 to 1.81 mg/L) than in free chlorine systems (+82% from 0.50 to 1.15 mg/L). While such trends may reflect the intended effects of the NAP, they may also coincide with internal system-level efforts or broader optimization initiatives not directly attributable to the state's NAP mandate. For instance, Louisiana issued a 0.5 mg/L minimum residual requirement during the same timeframe as the NAP mandate, whereas Texas implemented this years prior to the NAP. To verify this and distinguish NAP effects from these potential confounders, we first performed interrupted time series analysis combined with the difference-in-differences approach (ITS/DID). We also explored multilevel

A Louisiana



B Texas

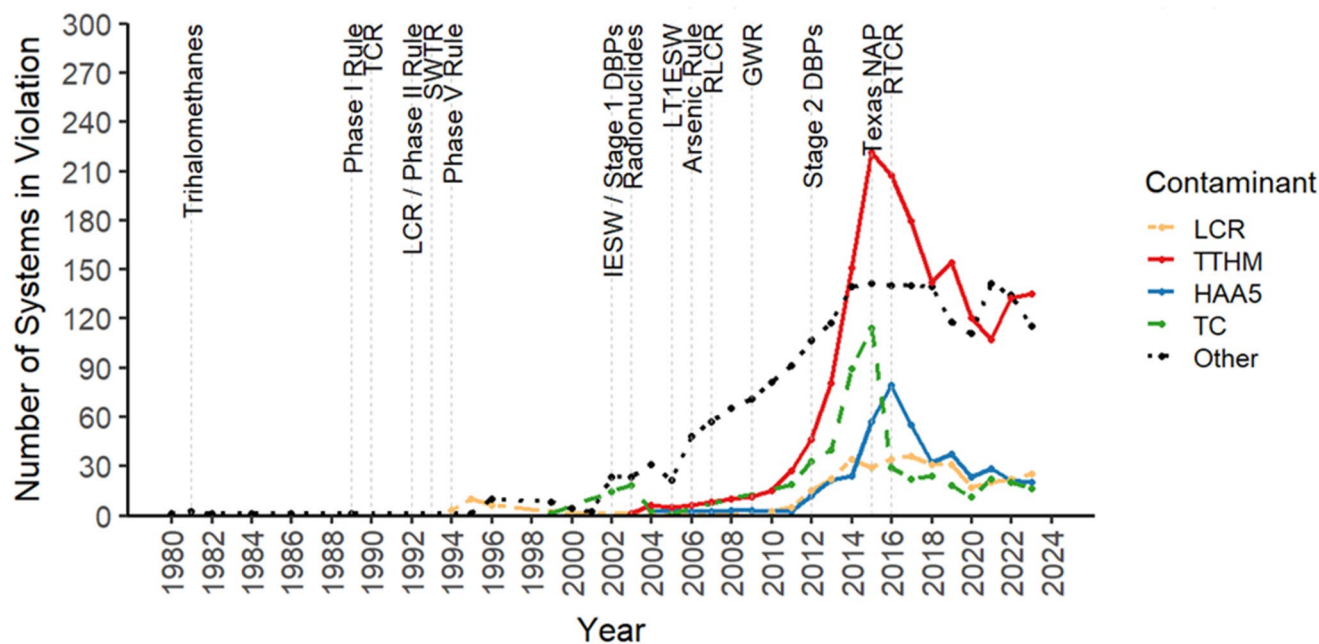


FIGURE 5 | Annual violation trends for (A) Louisiana (NAP enacted in 2014, enforced by 2017) and (B) Texas (NAP enacted and enforced in 2015). Each data point represents the number of PWSs with at least one reported violation in a given year, counted only once per system regardless of the number of violations. Vertical lines indicate the enforcement dates of major federal drinking water regulations since 1980.

modeling approaches (MLM), which allowed us to control for any unobserved heterogeneity across system types, states, and time periods.

Considering the large sample size (>1 million datapoints) and the proximity of the median to the mean, the subsequent analysis will examine the change in each contaminant's mean (average) value instead of the median. Given the more notable

increase and the known susceptibility of chloramine systems to nitrification, our analysis will also henceforth be focused primarily on chloramine systems.

To help interpret the case studies that follow, we briefly outline the two core analytical frameworks used: Difference-in-Differences (DID) and Interrupted Time Series (ITS). DID isolates the effects of NAP by comparing changes in treated

TABLE 2 | Chloramine systems: ITS effect estimates of NAP impact for each contaminant in Texas.

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Chloramine (mg/L as Cl_2)	Intercept (β_0)	2.09 (0.05)	***
	Time (β_1)	0.002 (0.005)	NS
	Intervention (β_2)	−0.04 (0.07)	NS
	Time since intervention (β_3)	0.02 (0.01)	**
HAA5 ($\mu\text{g/L}$)	Intercept (β_0)	18.75 (1.16)	***
	Time (β_1)	0.02 (0.12)	NS
	Intervention (β_2)	5.37 (1.47)	***
	Time since intervention (β_3)	−0.25 (0.15)	NS
TTHMs ($\mu\text{g/L}$)	Intercept (β_0)	34.80 (1.89)	***
	Time (β_1)	−0.04 (0.19)	NS
	Intervention (β_2)	1.94 (2.33)	NS
	Time since intervention (β_3)	−0.20 (0.23)	NS
Total coliform (% positive detects)	Intercept (β_0)	1.49 (0.14)	***
	Time (β_1)	0.02 (0.01)	*
	Intervention (β_2)	0.06 (0.19)	NS
	Time since intervention (β_3)	−0.04 (0.02)	**
HAA5 (% positive detects)	Intercept (β_0)	95.38 (0.28)	***
	Time (β_1)	−0.06 (0.03)	**
	Intervention (β_2)	1.05 (0.36)	***
	Time since intervention (β_3)	0.07 (0.04)	*
TTHMs (% positive detects)	Intercept (β_0)	0.96 (0.01)	***
	Time (β_1)	−0.001 (0.001)	*
	Intervention (β_2)	0.02 (0.01)	**
	Time since intervention (β_3)	0.001 (0.001)	NS

(Continues)

TABLE 2 | (Continued)

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Nitrate-N (mg/L)	Intercept (β_0)	1.15 (0.11)	***
	Time (β_1)	0.04 (0.01)	**
	Intervention (β_2)	−0.16 (0.14)	NS
	Time since intervention (β_3)	−0.05 (0.01)	**
Nitrite-N (mg/L)	Intercept (β_0)	0.03 (0.03)	NS
	Time (β_1)	−0.003 (0.003)	NS
	Intervention (β_2)	0.05 (0.04)	NS
	Time since intervention (β_3)	0.004 (0.004)	NS

^a $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.

systems (NAP states) to those in an external control group (non-NAP states), while ITS examines within-group trends over time using counterfactual projections. By combining both methods, the case studies provide a more robust evaluation of policy (i.e., NAP) impacts. As shown in Tables 3, 4, and 5, the DID framework is embedded in pre/post comparisons to adjust for broader temporal trends and contextual variation. For each analyte, we calculated changes before and after NAP adoption separately for Non-NAP (control) and NAP (intervention) states. The Non-NAP results reflect “business-as-usual trends”—what might have happened without any policy change—while the NAP results reflect both these background trends and any potential NAP-driven effects (see Figure S4). By subtracting the change in Non-NAP states from that in NAP states, the DID column isolates the adjusted policy impact while accounting for confounders.

3.3.1 | Case Study: Louisiana

This 2014–2017 timeline in Louisiana aligns with the state’s response to *Naegleria fowleri* infections, where the Louisiana Department of Health and Hospitals (LDHH) issued an Emergency Rule (ER) on November 6, 2013. A key component of this ER is the adoption of a mandatory minimum disinfectant residual of 0.5 mg/L (either free or total chlorine) to apply across the distribution system, including finished water storage tanks. Prior to this, Louisiana did not enforce a numerical threshold for minimum chlorine residuals in drinking water. Instead, systems were required only to maintain a “trace” or detectable level of free chlorine residual throughout the system (Louisiana Department of Health 2013). Another key component, as outlined in LAC 51:XII §367, *Disinfectant Residual Monitoring and Record Keeping*, was the requirement for utilities to submit NAPs starting on March 1, 2014, and requiring all systems to submit

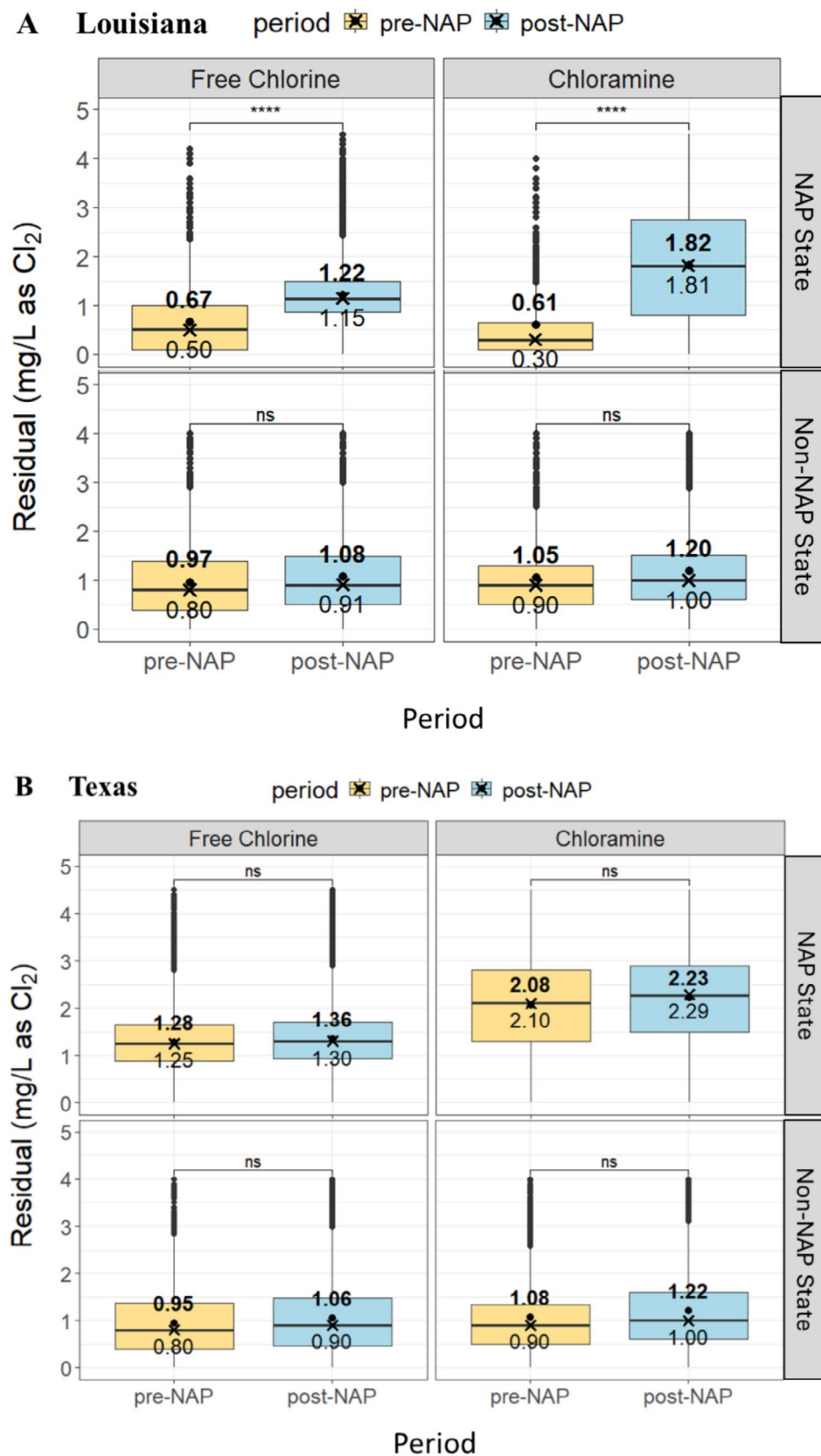


FIGURE 6 | Boxplots depict disinfectant residual concentrations (mg/L as Cl₂) before and after NAP implementation for (A) Louisiana: NAP date = 2014 and (B) Texas NAP date = 2015, faceted by disinfectant type (free chlorine, chloramine) and NAP status (NAP States vs. Non-NAP States in EPA Region 6). We included the median values (non-bold number, horizontal line on the boxplot) and the average (bolded number). See Figure S6 for corresponding quarterly trends.

TABLE 3 | Chloramine systems: ITS/DID treatment differences in mean and standard deviation (S.D.) based on Louisiana's Nitrification Action Plan enforcement date (2009–2019).

Contaminant	Non-NAP ^a					NAP (Louisiana) ^b					Diff-in-Diff ^c	
	(# unique PWS ^d = 1804)					(# unique PWS ^d = 149)					Adjusted	
	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Δ^e	p^g
Chloramine (mg/L as Cl ₂)	1.05 (0.08)	1.21 (0.06)	0.16	0.16, 0.16	***	0.61 (0.27)	1.82 (0.15)	1.21	1.20, 1.20	***	1.05	***
HAA5 (µg/L)	19.70 (3.98)	28.17 (4.24)	8.47	7.90, 9.00	***	19.00 (2.66)	25.41 (4.33)	6.41	6.30, 6.50	***	-2.06	***
TTHMs (µg/L)	44.21 (10.39)	51.33 (5.33)	7.12	5.80, 8.50	***	32.82 (5.33)	34.37 (4.34)	1.55	1.40, 1.70	NS	-5.57	***
Total coliform (% detects)	0.95 (0.35)	0.94 (0.31)	-0.01	-0.01, -0.01	NS	28.47 (4.02)	2.54 (6.04)	-25.93	-26, -26	***	-25.99	***
HAA5 (% detects)	86.04 (6.75)	90.48 (7.87)	4.44	3.50, 5.40	***	87.44 (3.41)	93.15 (4.42)	5.71	5.60, 5.90	***	1.27	***
TTHMs (% detects)	96.68 (1.91)	96.26 (2.97)	-0.42	-0.72, -0.12	NS	96.04 (3.47)	95.00 (2.60)	-1.04	-1.20, -0.92	NS	-0.58	**

^aStates without NAP within EPA Region 6—New Mexico, Oklahoma, and Arkansas—analyzed based on Louisiana's NAP date.^bVoluntary implementation date: 2014; mandatory compliance date: 2017.^cDifference-in-Difference ($\Delta_{adj.} = \Delta_{Non-NAP} - \Delta_{Non-NAP}$), which is the causal effect of NAP impact.^dOnly include chloraminated community water systems (CWS) with distribution system (DS) data.^eWelch two sample *t*-test.^fCI = confidence interval.^g* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.**TABLE 4** | Free chlorine systems: ITS/DID treatment differences in mean and standard deviation (S.D.) based on Louisiana's Nitrification Action Plan enforcement date (2009–2019).

Contaminant	Non-NAP ^a					NAP (Louisiana) ^b					Diff-in-Diff ^c	
	(# unique PWS ^d = 1669)					(# unique PWS ^d = 954)					Adjusted	
	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Δ^e	p^g
Free Chlorine (mg/L as Cl ₂)	0.96 (0.09)	1.04 (0.06)	0.08	0.08, 0.08	***	0.67 (0.10)	1.22 (0.07)	0.55	0.55, 0.55	***	0.47	***
HAA5 (µg/L)	31.30 (5.95)	30.61 (3.68)	0.69	-1.3, -0.04	*	10.13 (4.03)	12.21 (4.19)	2.1	1.9, 2.2	***	1.41	***
TTHMs (µg/L)	52.56 (8.11)	52.31 (7.24)	-0.24	-1.2, 0.76	NS	23.79 (6.98)	28.46 (6.67)	4.7	4.4, 4.9	NS	4.94	***
Total coliform (% detects)	1.07 (0.55)	0.93 (0.42)	-0.14	-0.14, -0.14	***	25.51 (3.66)	2.83 (5.84)	-23	-23, -23	***	-22.86	***
HAA5 (% detects)	79.84 (7.87)	88.86 (17.27)	-6.9	-8.2, -5.7	***	62.19 (8.83)	78.23 (13.49)	16	16, 16	***	22.9	***
TTHMs (% detects)	92.63 (5.24)	91.55 (7.18)	-3.8	-4.5, -3.0	NS	82.85 (6.60)	88.89 (5.05)	6.0	5.8, 6.2	NS	9.8	***

^aStates without NAP within EPA Region 6—New Mexico, Oklahoma, and Arkansas—analyzed based on Louisiana's NAP date.^bVoluntary implementation date: 2014; mandatory compliance date: 2017.^cDifference-in-Difference ($\Delta_{adj.} = \Delta_{Non-NAP} - \Delta_{Non-NAP}$), which is the causal effect of NAP impact.^dOnly include chloraminated community water systems (CWS) with distribution system (DS) data.^eWelch two sample *t*-test.^fCI = confidence interval.^g* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.

TABLE 5 | Adjusted estimates of chloramine residual changes following NAP implementation in Louisiana.

Contaminant	Diff-in-Diff ^a	Diff-in-Diff ^b	Diff-in-Diff ^c	Diff-in-Diff ^d	<i>p</i> ^e
	Raw change Δ^e	Adjusted change by Non-NAP control Δ^e	Adjusted change by free Cl ₂ control Δ^e	Adjusted change by adjusted free Cl ₂ control Δ^e	
Chloramine (mg/L as Cl ₂)	1.21	1.05	0.66	0.58	***
HAA5 (µg/L)	6.41	−2.06	4.31	5.00	***
TTHMs (µg/L)	1.55	−5.57	−3.15	−3.39	***
Total coliform (% detects)	−25.93	−25.99	−2.93	−3.07	***
HAA5 (% detects)	5.71	1.27	−10.29	−17.19	***
TTHMs (% detects)	−1.04	−0.58	−7.04	−10.84	***

^aSee Table 3, Louisiana (NAP) column. Raw change reflects the difference between post- and pre-intervention values based on counterfactual trends within the group (i.e., without using a control group). This is unadjusted.

^bSee Table 3, Diff-in-Diff (DID) adjusted column. Values reflect changes in NAP systems adjusted using trends from non-NAP systems.

^cSee Tables 3 and 4. This value represents the raw change in chloramine systems in Louisiana, adjusted by subtracting the change observed in free chlorine systems from the same state.

^dSee Tables 3 and 4. This value is the DID-adjusted change for chloramine systems further adjusted by subtracting the free chlorine adjusted change, isolating the net NAP effect specific to chloramine systems.

^e**p* < 0.05, ***p* < 0.01, ****p* < 0.001, NS = not significant. All Diff-in-Diff were tested separately; all reported ****p* values.

NAPs no later than January 1, 2017 (Louisiana Department of Health 2016). The implementation of the ER, which included the NAP requirement among other measures, led to more significant improvements in water quality. This distinction in implementation dates provides a critical context for our comparative analysis of the impact of NAPs on water quality across these different states.

Results in Table 3 showed there was an overall increase in the mean chloramine residual concentrations for Louisiana from 0.61 mg/L-Cl₂, which is below the regional average of 1.05 mg/L-Cl₂, to 1.82 mg/L-Cl₂, exceeding the regional average. This increase coincided with a 26% reduction in positive total coliform detections, suggesting that maintaining residuals above 1 mg/L may help reduce microbial presence. However, higher residuals may involve tradeoffs, as some systems also exhibited modest increases in HAA5 and TTHM concentrations. Still, statistical significance does not always capture practical or health-relevant impacts. For example, while HAA5 concentrations in NAP states significantly increased by 6.41 µg/L in raw terms, the DID-adjusted change actually showed a small significant decrease of −2.06 µg/L. At the same time, HAA5 occurrences increased by 5.71% within-state and 1.27% after adjustment. This suggests that although HAA5 levels increased over time within the state, similar or larger increases occurred in states without NAPs. In other words, the apparent increase may not be attributable to NAP implementation but rather to broader regional factors. The adjusted decrease implies that, relative to regional trends, NAP states may have actually experienced better control of HAA5. Similarly, for TTHMs, while concentrations rose slightly from 32.82 to 34.37 µg/L (+1.55 µg/L) in raw within-state comparisons, this modest increase may be arguably less concerning given the 1% decline in detection frequency.

Shifting from DID to ITS, we assess not just whether NAP had an impact, but when those effects occurred. The DID showed a statistically significant increase in chloramine residuals across both NAP and Non-NAP states, but the effect was substantially higher in NAP states (+1.21 vs. +0.16 mg/L). This suggests that while both groups saw increases, the shift was overall more meaningful in systems with NAP in place. The time-varying ITS model results (Table 6), which use within-group counterfactuals, reinforce this. The strongest chloramine increase occurred immediately following NAP implementation ($\beta_2 = 0.77$, $p < 0.001$), suggesting that the mandate may have effectively boosted disinfectant residuals across the distribution system. This increase appears stable over time, with little additional change observed ($\beta_3 = 0.03$, $p = \text{NS}$). In contrast, HAA5 detection shows no immediate response to NAP implementation ($\beta_2 = 0.24$, $p > 0.05$), but a significant upward trend emerges over time ($\beta_3 = 0.55$, $p < 0.001$), indicating a gradual rather than abrupt change. In contrast, HAA5 concentrations spike immediately post-intervention ($\beta_2 = 5.50$), but then declined gradually, with no statistically significant trend over time ($\beta_1 = -0.03$, $\beta_3 = 0.16$, $p = \text{NS}$). Meanwhile, the percentage of positive HAA5 detections did not increase immediately ($\beta_3 = -1.75$, $p = \text{NS}$), but rose significantly over time ($\beta_3 = 0.55$, $p < 0.001$). This suggests a potential tradeoff with an initial spike in concentration followed by a gradual decline to stable levels, while the frequency of detections may continue to rise, reflecting more consistent but not necessarily higher concentration over time. The 26% reduction in TC suggests that NAP implementation may support improved microbial control. This effect has a delay of approximately 3 years (see Figure S5b) after the NAP mandate, reflected by a significant decline in total coliform levels over time ($\beta_3 = -1.67$). This 3-year lag for TC reduction may be due to the phased rollout of the policy, with voluntary compliance starting in 2014 and full implementation required by 2017. Tables 4 and 7 provide results for free chlorine systems, which can also serve as additional control to help

TABLE 6 | Chloramine systems: ITS effect estimates of NAP impact for each contaminant in Louisiana.

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Chloramine (mg/L as Cl_2)	Intercept (β_0)	0.60 (0.14)	***
	Time (β_1)	0.003 (0.01)	NS
	Intervention (β_2)	0.77 (0.19)	***
	Time since intervention (β_3)	0.03 (0.02)	NS
HAA5 ($\mu\text{g/L}$)	Intercept (β_0)	18.51 (1.78)	***
	Time (β_1)	−0.03 (0.15)	NS
	Intervention (β_2)	5.50 (2.43)	**
	Time since intervention (β_3)	0.16 (0.21)	NS
TTHMs ($\mu\text{g/L}$)	Intercept (β_0)	32.16 (2.36)	***
	Time (β_1)	−0.04 (0.20)	NS
	Intervention (β_2)	0.28 (3.22)	NS
	Time since intervention (β_3)	0.20 (0.28)	NS
Total coliform (% positive detects)	Intercept (β_0)	26.89 (1.53)	***
	Time (β_1)	−0.15 (0.13)	NS
	Intervention (β_2)	8.21 (2.11)	***
	Time since intervention (β_3)	−1.67 (0.18)	***
HAA5 (% positive detects)	Intercept (β_0)	87.39 (0.85)	***
	Time (β_1)	−0.01 (0.07)	NS
	Intervention (β_2)	0.24 (1.16)	NS
	Time since intervention (β_3)	0.55 (0.10)	***
TTHMs (% positive detects)	Intercept (β_0)	0.95 (0.01)	***
	Time (β_1)	−0.001 (0.001)	*
	Intervention (β_2)	−0.01 (0.01)	NS
	Time since intervention (β_3)	0.002 (0.001)	**

(Continues)

TABLE 6 | (Continued)

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Nitrate–Nitrite (mg/L as N) ^b	Intercept (β_0)	0.03 (0.10)	NS
	Time (β_1)	−0.02 (0.01)	**
	Intervention (β_2)	0.26 (0.13)	*
	Time since intervention (β_3)	0.01 (0.01)	NS

^a* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.^bNot enough distribution system nitrate and nitrite data are available to perform meaningful analysis.

distinguish chloramine-specific effects. Nitrate–nitrite data is included; however, the limited number of distribution system samples makes it difficult to apply ITS or DID analyses.

Since NAP policies are designed to improve chloraminated systems, changes observed in free chlorine systems (Table 5 and Figure 4) likely reflect broader state or regional trends, including the coinciding 0.5 mg/L minimum residual requirement, rather than the policy itself. Comparing chloramine and free chlorine trends allows for better attribution of observed effects. In Louisiana, chloramine systems saw a 1.21 mg/L- Cl_2 increase in residual, while free chlorine systems increased by 0.55 mg/L- Cl_2 . Subtracting this shared baseline shift yields an adjusted increase of 0.66 mg/L- Cl_2 , more plausibly attributed to NAP-specific efforts. For a more conservative adjustment, we also accounted for changes observed in both free chlorine systems within Louisiana and in non-NAP states. Table 5 compares different methods for estimating the effect of NAP policies on chloramine residuals. We report several estimates to isolate the effect of NAP on chloraminated systems: raw pre/post changes, difference-in-differences estimates using non-NAP states as external controls, adjustments using free chlorine systems as internal controls, and a combined adjustment. This approach strengthens the evidence that NAP specifically improved residual maintenance in its intended targets. See footnotes a–d for calculation details.

3.3.2 | Case Study: Texas

Results from ITS/DID in Table 5 showed that during the post-intervention period, chloramine residual in Texas had an average value of approximately 2.23 mg/L. In the absence of an intervention, an average response of 2.06 mg/L would be expected. These observed values are much higher than the regional average of 1.08 mg/L Cl_2 . Texas results suggest the state may actually outperform non-NAP states even prior to NAP implementation, so the observed impact of NAP shown in Table 5 appears smaller than in states with lower disinfectant residuals like Louisiana. This could reflect that many Texas utilities were already following the best practices recommended by NAP before it became an official requirement. In fact, Texas enforced a minimum residual

TABLE 7 | Free chlorine systems: ITS effect estimates of NAP impact for each contaminant in Louisiana.

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Chloramine (mg/L as Cl_2)	Intercept (β_0)	0.59 (0.04)	***
	Time (β_1)	−0.01 (0.003)	*
	Intervention (β_2)	0.19 (0.06)	***
	Time since intervention (β_3)	0.03 (0.005)	***
HAA5 ($\mu\text{g/L}$)	Intercept (β_0)	10.76 (1.86)	***
	Time (β_1)	−0.27 (0.16)	*
	Intervention (β_2)	−1.75 (2.54)	NS
	Time since intervention (β_3)	0.67 (0.22)	***
TTHMs ($\mu\text{g/L}$)	Intercept (β_0)	27.92 (3.34)	***
	Time (β_1)	−0.13 (0.28)	NS
	Intervention (β_2)	−1.02 (4.55)	NS
	Time since intervention (β_3)	0.52 (0.39)	NS
Total coliform (% positive detects)	Intercept (β_0)	23.88 (2.27)	***
	Time (β_1)	−0.15 (0.13)	NS
	Intervention (β_2)	8.21 (2.11)	***
	Time since intervention (β_3)	−1.67 (0.18)	***
HAA5 (% positive detects)	Intercept (β_0)	65.08 (3.90)	***
	Time (β_1)	−0.28 (0.33)	NS
	Intervention (β_2)	−4.79 (5.32)	NS
	Time since intervention (β_3)	2.09 (0.46)	***
TTHMs (% positive detects)	Intercept (β_0)	0.88 (0.02)	***
	Time (β_1)	0.001 (0.002)	NS
	Intervention (β_2)	0.01 (0.03)	NS
	Time since intervention (β_3)	0.001 (0.003)	NS

(Continues)

TABLE 7 | (Continued)

Characteristic	Dependent variables		
	Coefficient	Effect (S.D.)	p^a
Nitrate (mg/L as N)	Intercept (β_0)	0.35 (1.40)	NS
	Time (β_1)	−0.02 (0.12)	NS
	Intervention (β_2)	−0.68 (1.65)	NS
	Time since intervention (β_3)	0.10 (0.14)	NS
Nitrite (mg/L as N)	Intercept (β_0)	−1.80E−17 (0.05)	NS
	Time (β_1)	−1.90E−18 (0.004)	NS
	Intervention (β_2)	−0.05 (0.05)	NS
	Time since intervention (β_3)	0.01 (0.005)	NS
Nitrate–Nitrite (mg/L as N)	Intercept (β_0)	0.08 (0.01)	***
	Time (β_1)	−0.001 (0.001)	NS
	Intervention (β_2)	−0.01 (0.02)	NS
	Time since intervention (β_3)	0.004 (0.002)	**

^a $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.

requirement well before adopting NAP, with utilities falling below 0.5 mg/L subject to boil water orders (30 Tex. Admin. Code § 290.46(q)(6)(A)(ii), adopted Sept. 13, 2000). To avoid such orders, most chloraminating utilities already maintained higher residuals across their systems. In this way, the adoption of NAP may be intertwined with “learning by doing” by state regulators and utility personnel who had already built capacity and procedures aligned with NAP goals. Although data availability is limited, a similar ITS analysis of distribution system nitrate-N concentrations suggests a gradual post-NAP decline. While the immediate change was not significant ($\beta_2 = -0.16$, $p = \text{NS}$), the negative slope over time ($\beta_3 = -0.05$, $p < 0.01$) indicates a downward trend that may reflect improved nitrification control practices. Changes in nitrite-N concentrations were not statistically significant.

Table 8 reports a 0.17 mg/L increase in chloramine residual (+0.03 mg/L DID adjusted) observed alongside a 3.01 $\mu\text{g/L}$ increase in HAA5 concentrations (−5.16 $\mu\text{g/L}$ DID adjusted). There is no significant changes in the probability of TC, HAA5, and TTHMs detection post-NAP. In fact, when accounting for differences between Texas and non-NAP states, the percentage of detections for TC, HAA5, and TTHMs in Texas appears lower than those in EPA Region 6. The ITS effect (slope) in Table 8 showed that NAP implementation had no significant immediate impact on chloramine residual (β_2

TABLE 8 | Chloramine systems: ITS/DID treatment differences in mean and standard deviation (S.D.) based on Texas's Nitrification Action Plan enforcement date (2010–2020).

Contaminant	Non-NAP ^a					NAP (Texas) ^b					Diff-in-Diff ^c	
	(# unique PWS ^d = 1777)					(# unique PWS ^d = 3443)					Adjusted	
	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Pre-NAP	Post-NAP	Δ^e	95% CI ^{e,f}	p^g	Δ^e	p^g
Chloramine (mg/L as Cl_2)	1.08 (0.06)	1.22 (0.04)	0.14	0.14, 0.14	***	2.06 (0.09)	2.23 (0.13)	0.17	0.17, 0.17	***	0.03	***
HAA5 (μ g/L)	21.27 (4.01)	29.44 (3.68)	8.17	7.80, 8.60	***	18.67 (1.82)	21.68 (2.61)	3.01	3.00, 3.00	***	-5.16	***
TTHMs (μ g/L)	45.26 (9.03)	50.76 (5.42)	5.50	4.60, 6.40	***	34.93 (3.73)	34.21 (2.92)	-0.72	-0.78, -0.66	NS	-6.22	***
Total coliform (% detects)	0.92 (0.34)	1.02 (0.42)	0.11	0.10, 0.11	NS	1.32 (0.56)	1.36 (0.49)	0.03	0.03, 0.04	NS	-0.08	***
HAA5 (% detects)	84.60 (8.42)	92.56 (5.36)	7.96	7.10, 8.80	***	95.75 (1.06)	96.53 (0.80)	0.78	0.77, 0.80	NS	-7.18	***
TTHMs (% detects)	95.66 (3.84)	96.48 (2.06)	0.82	0.43, 1.20	NS	96.25 (2.89)	97.17 (0.68)	0.92	0.88, 0.96	NS	0.10	NS

^aStates without NAP within EPA Region 6—New Mexico, Oklahoma, and Arkansas—analyzed based on Texas's NAP date.^bVoluntary implementation date: 2015; mandatory compliance date: 2015.^cDifference-in-difference ($\Delta_{adj} = \Delta_{Non-NAP} - \Delta_{NAP}$), which is the causal effect of NAP impact.^dOnly include chloraminated community water systems (CWS) with distribution system (DS) data.^eWelch Two Sample t -test.^fCI = confidence interval.^g* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.

slope $\cong 0$, $p = NS$) which suggests that states with higher residual prior to the NAP date will not gain any substantial benefit from mandating a NAP in 2015. It is likely that utilities in Texas were already proactively monitoring residuals and controlling nitrification prior to this NAP implementation date, or that their operations more closely resemble those of non-NAP states in the region.

3.3.3 | Disentangling Policy Effects: A Multilevel Approach

While no substantial increase in free chlorine residuals was observed in non-NAP states, Louisiana did experience an increase in both chloramine and free chlorine systems. This raised questions about whether NAP policies also affected free chlorine systems, whether observed changes in chloraminated systems can be attributed to NAP, or whether other factors were involved. One possibility is that Louisiana's response was influenced by *N. fowleri* concerns; however, this remains uncertain, as other EPA Region 6 states (e.g., Oklahoma, Arkansas, Texas) reportedly faced similar risks and may have adopted parallel but informal strategies (Hall et al. 2024; Ward and Sherchan 2023; Yoder et al. 2010). To investigate whether the observed increase in chloramine residuals in Table 5 reflects true NAP-driven effects rather than broader statewide operational changes, we applied a multilevel model. This model estimates policy effects that may vary by time (pre/post), policy exposure (NAP vs. non-NAP), and disinfection type (chloramine vs. free chlorine).

The final multilevel model includes three hierarchical levels to index observations across public water systems (PWSs, j), time points (t), and policy group/disinfection type combinations (i). It incorporates four-way interactions among time, policy, policy group, and disinfection type (DISTYPE), with random intercepts for each PWSID. This structure allows us to isolate NAP-specific effects within targeted subgroups (e.g., NAP chloramine systems over time). This final model below was selected based on the lowest BIC value among the candidate models (see Table S5 for details).

Final Model Equation (Model 8):

$$\begin{aligned} \text{Value}_{it} = & \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{Policy}_j + \beta_3 \text{PolicyGroup}_i \\ & + \beta_4 \text{DISTYPE}_i + \beta_5 (\text{Time}_t \times \text{Policy}_j) + \dots \\ & + \beta_{15} (\text{Time}_t \times \text{Policy}_j \times \text{PolicyGroup}_i \times \text{DISTYPE}_i) + u_j + \epsilon_{ij} \end{aligned}$$

where β_0 : overall intercept (grand mean); β : effects unique to each variable; Value_{ij} : outcome (e.g., disinfectant residual) for system i at observation (e.g., time) j ; Time_t : continuous time (e.g., by quarter); Policy_j : Pre-NAP (=1), Post-NAP (=0); PolicyGroup_i : NAP state (=1), Non-NAP state (=0); DISTYPE_i : Chloramine (=1), Free chlorine (=0); $u_{ij|PWSID_i}$: random intercept for group PWSID (e.g., random variation across PWSID), assumed to follow $u_{ij|PWSID_i} \sim N(0, \sigma_u^2)$. N stands for “normally distributed”; 0 is the mean distribution where we assume the random effect has an average of zero; σ_u^2 is the variance of the random effect; ϵ_{it} : residual error for each observation, assumed $\sim N(0, \sigma_u^2)$.

TABLE 9 | Final multilevel model specification selected based on the lowest BIC value among all candidate models.

Model formula: $\text{value} \sim \text{time} \times \text{policy} \times \text{policygroup} \times \text{DISTYPE} + (1 \text{PWSID})$								
Model	Fixed effects term	Estimate	std.error	p	conf.low	conf.high	BIC	ICC
8	(Intercept)***	0.6821	0.0098	0.0000	0.6628	0.7014	418,636	0.71891
	Time***	0.0022	0.0001	0.0000	0.0019	0.0024		
	Policy***	0.0939	0.0026	0.0000	0.0889	0.0989		
	Policygroup	−0.0003	0.0199	0.9896	−0.0392	0.0387		
	DISTYPE***	0.1236	0.0026	0.0000	0.1185	0.1287		
	Time:policy	−0.0002	0.0002	0.4558	−0.0006	0.0003		
	Time:policygroup***	−0.0059	0.0007	0.0000	−0.0074	−0.0045		
	Policy:policygroup***	0.2508	0.0107	0.0000	0.2298	0.2717		
	Time:DISTYPE***	−0.0031	0.0002	0.0000	−0.0035	−0.0028		
	Policy:DISTYPE***	−0.0334	0.0030	0.0000	−0.0392	−0.0276		
	Policygroup:DISTYPE	0.0184	0.0346	0.5957	−0.0495	0.0863		
	Time:policy:policygroup***	0.0168	0.0008	0.0000	0.0152	0.0185		
	Time:policy:DISTYPE***	0.0036	0.0003	0.0000	0.0031	0.0041		
	Time:policygroup:DISTYPE	0.0034	0.0018	0.0587	−0.0001	0.0069		
	Policy:policygroup:DISTYPE***	0.2336	0.0246	0.0000	0.1854	0.2819		
	Time:policy:policygroup:DISTYPE	−0.0023	0.0019	0.2384	−0.0061	0.0015		

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, NS = not significant.

Key findings in Table 9 showed several significant patterns. First, there is a slight overall increase in disinfectant residual value over time ($+0.0022$, $p < 0.001$), and levels are significantly higher post-policy ($+0.0939$, $p < 0.001$). Chloramine systems also have a higher baseline compared to free chlorine systems ($+0.1236$, $p < 0.001$). The DID effect for *policy* × *policy_group* ($+0.2508$, $p < 0.001$) suggests the disinfectant residual levels increased more in NAP states post-policy than expected based on trends in non-NAP states. The interactions *time* × *policy* ($−0.0002$, $p = 0.4558$), *policy* × *disinfection_type* ($−0.0334$, $p < 0.001$), and *time* × *disinfection_type* ($−0.0031$, $p < 0.001$) showed a negative effect. However, these trends should be interpreted with caution, as models excluding *policy_group* combine all chloramine systems, where non-NAP systems greatly outnumber NAP systems, potentially diluting or masking NAP-specific effects.

Focusing specifically on chloramine systems in NAP states, the three-way interaction *policy* × *policy_group* × *disinfection_type* shows a strong positive effect ($+0.2336$, $p < 0.001$), supporting the conclusion that chloramine systems in NAP states saw a substantial positive change post-policy beyond what was observed in free chlorine or non-NAP states. Finally, the effects of the interaction *time* × *policy* × *policy_group* ($+0.0168$, $p < 0.001$) indicate continued post-policy improvement over time in NAP states. In contrast, *policy_group* alone ($−0.0003$, $p = 0.9896$) was not significant, indicating no baseline difference between NAP and non-NAP states. Other higher-order interactions, including the four-way interaction (*time* × *policy* × *policy_group* × *disinfection_type*), were also not significant, suggesting the full four-way term may not be necessary.

These MLM findings largely align with those from the ITS and DID analyses presented earlier. The model fit was strong (BIC = 418,646), and the intraclass correlation coefficient (ICC = 0.72) indicates that a large proportion of variance is attributable to differences across PWSIDs. This high ICC supports the use of a multilevel model, reflecting meaningful system-level variation likely driven by shared regulatory environments, climatic conditions, and operational practices within groups.

Current data limitations restrict a full evaluation of NAP impacts on nitrification and distribution system water quality. Therefore, many analyses rely on indicators such as residual disinfectant levels and microbial detections. Future work should focus on obtaining more data directly from utilities and building a larger database that maps seller-purchaser utility relationships, allowing the use of point-of-entry data from purchased systems as a surrogate for distribution system quality. With a more comprehensive and standardized dataset, multilevel models can be expanded to account for contextual factors at the EPA region, State, and PWS levels, enabling a more generalizable assessment evaluation of how NAPs influence nitrification control and distribution water quality nationwide.

4 | Conclusions

We demonstrated how controlled interrupted time series analyses, difference-in-differences, and multilevel modeling can be used to evaluate the effectiveness of Nitrification Action Plans and their potential tradeoffs. These approaches help answer

important questions: *Do the most vulnerable states benefit from regulation? Are there unintended consequences, such as increased disinfection byproduct formation or LCR violations?* Using ITS, DID, and MLM, we found strong evidence that NAP policies improve disinfectant residual maintenance specifically in chloraminated systems. The multilevel model, which clusters by PWSID, confirmed this effect through a significant three-way interaction between policy, policy group (NAP vs. non-NAP), and disinfection type. This interaction shows that residuals increased more in chloraminated NAP systems than free chlorine or non-NAP systems. Such findings underscore the value of hierarchical models like MLM in isolating effects that could otherwise be obscured by system-level changes and cause bias.

In our analysis, free chlorine systems and non-NAP states served as controls for confounding factors such as regulatory agency policies and procedures or other temporal changes that coincided with NAP implementation. Results from pre and post intervention periods demonstrate that a state with nitrification issues can benefit greatly from state regulation as the NAP states reported an increase in distribution system residual without seeing any substantial increase in DBP, TC, and LCR violations and occurrences long term. Maintaining chloramine residuals ideally targeting 1.5 mg/L-Cl₂ is consistent with recommendations from the AWWA M56 manual and is a common target included in many NAPs. This disinfectant residual target has been associated with improved microbial control, including significant reductions in total coliform and potentially other microbial growth. It may also help mitigate the risk of nitrification, especially when paired with proactive monitoring and response strategies outlined in a NAP. These findings are informative for policymakers, raising important considerations and demonstrating that NAPs may help maintain disinfectant residuals and reduce total coliform detections. However, potential tradeoffs with DBP formation should be carefully considered. Understanding how and when changes occur, and whether they are meaningful in practice, is key to interpreting NAP's effects on drinking water quality.

Considering the date of NAP implementation can vary from the deadline for compliance, delays in the anticipated effects may occur. These examples highlight the importance of pairing statistical results with real-world context. A statistically significant result is not always practically significant, and a non-significant finding does not necessarily mean there is no impact. As shown, effects may be subtle, delayed, masked by background variation, or, alternatively, the influence of broader operational trends seen in non-NAP systems. States with a small proportion of chloraminated systems and minimal nitrification issues may not see the need to develop a plan; furthermore, requiring utilities to have a plan can impose a burden on small systems with limited manpower and resources. This is often viewed as one of the significant challenges. However, in states grappling with significant chlorine decay and nitrification concerns, implementing such a plan has proven to be promising. It offers struggling systems the opportunity to enhance their water quality through offering a promising tool to improve water quality through structured guidance, monitoring programs, and training.

Author Contributions

Natchaya Luangphairin: data curation, formal analysis, visualization, methodology, writing – original draft. **J. Alan Roberson:** conceptualization, supervision, funding acquisition, writing – review and editing. **Katherine Alfredo:** conceptualization, resources, supervision, methodology, writing – review and editing.

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Conflicts of Interest

J. Alan Roberson is the Executive Director of ASDWA, the professional organization that provided funding that supported this research. This affiliation did not influence the study design, data collection, analysis, or interpretation.

Data Availability Statement

Data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.